

S. KASINADH SARMA

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SUMMARY

Hi, I'm **Kasinadh Sarma** — a Software Engineer with a B.Tech in Cyber Security, specializing in **Application Security (AppSec)** and **DevSecOps** within production web environments. Proven ability to integrate security into the SDLC by **hardening API routes**, enforcing strict **secrets hygiene** in CI/CD pipelines, and maintaining **zero-vulnerability dependencies**. Experienced in securing **headless CMS** architectures, integrating payment gateways (**Razorpay**) with robust input validation, and implementing safe file-handling practices for medtech (**ROPODS**) and AI platforms. Fluent in leveraging **Generative AI tools** (Claude, Gemini, ChatGPT) to accelerate secure development, automate code review, and prototype AI-augmented engineering workflows.

EXPERIENCE

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| Junior Engineer | ROPODS (Medtech Startup), Bengaluru | <i>06/2025 – Present</i> |
|------------------------|--|--------------------------|
- Full-stack engineer across corporate web, e-commerce, and device-adjacent tooling, with a security-first approach to configuration and API design.*
- **E-Commerce Security & API Hardening:** Secured the shop.ropods.com storefront and **Razorpay** payment integration; engineered internal API routes with rigorous **input validation** and **Role-Based Access Control (RBAC)** checks, preventing injection vulnerabilities and securing checkout flows.
 - **Bot Mitigation & Endpoint Security:** Engineered a **dual-layer CAPTCHA** system across all public-facing endpoints (Contact Us, Book Demo) on the corporate site, integrating **Google reCAPTCHA v3** alongside a custom math-based logic challenge to thwart automated bot attacks, form spam, and abuse vectors.
 - **DevSecOps & CI/CD:** Architected secure **Next.js** deployment pipelines on **Vercel** via **GitHub Actions**; enforced strict environment-scoped secrets management and tightened **.env/deployment-key** handling, eliminating credential exposure risks across production environments.
 - **Dependency Auditing:** Drove continuous static analysis (**SAST**) and dependency auditing across corporate and e-commerce properties, maintaining a baseline of zero known high-severity CVEs in **node_modules** and resolving security-related TypeScript/ESLint errors.
 - **Protocol Research:** Investigated **UCP** (Universal Commerce Protocol) and **ACP** (Agentic Commerce Protocol) standards to inform secure platform-to-device communication interfaces for IoT/medtech integrations.
 - **AI-Assisted Development:** Leveraged **Claude** and **ChatGPT** to accelerate secure code authoring, RCA documentation, and QA test-case generation, keeping manual security review as the final gate before merge.

TRAINING & INTERNSHIPS

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| Summer Trainee | Intel Unnati | <i>05/2023 – 07/2023</i> |
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- Built an **AI-powered road object detection** system with a GTK GUI using the **COCO dataset** and **OpenCV**, implementing **memory-safe image preprocessing** and strict input validation to mitigate buffer overflows and adversarial input vectors. (**GitHub**)
 - Designed secure data ingestion pipelines for large-scale **TensorFlow** image datasets, ensuring data integrity, confidentiality, and tamper-proofing throughout the model training lifecycle.

PROJECTS

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|---------------|---------------------------------|
| CIELON | <i>Freelance Client Project</i> |
|---------------|---------------------------------|
- **Backend Architecture:** Architected the secure backend infrastructure for a greenfield **Next.js 16** storefront from scratch, enforcing strict **TypeScript** type-safety across all server-side functions to prevent runtime injection attacks and secure API endpoints.
 - **Content Synchronization Pipeline:** Built a secure backend content synchronization pipeline between **Sanity Studio** and **Firestore**; implemented a custom read cache alongside strict **Firestore Security Rules** to prevent unauthorized data mutations and block unauthenticated API access.
 - **Security Documentation:** Authored comprehensive backend security architecture documentation and database schema designs, focusing on data confidentiality and **Zero Trust principles** to keep Firestore strictly isolated from client-side manipulation.

SKILLS

- **Security & AppSec:** API Hardening, Input Validation, Secrets Hygiene, Dependency Auditing, RBAC
- **Cloud & DevSecOps:** GCP, Firebase Security Rules, Vercel, GitHub Actions CI/CD Security
- **Languages:** Python, TypeScript, Dart, Bash
- **Frontend & Backend:** Next.js, React, Node.js, REST APIs
- **Generative AI Tools:** Claude (Code, API), Gemini, ChatGPT, Prompt Engineering, AI-Assisted Secure Code Review
- **AI/ML Security:** Memory-Safe Preprocessing, Secure Data Ingestion (TensorFlow, OpenCV)
- **Tools:** Git, VS Code, Linux

EDUCATION

Parul University

B.Tech in Computer Science and Engineering (Cyber Security)

CGPA: 7.46

Vadodara

November 2020 – March 2024

CERTIFICATIONS

- TryHackMe Sec0 Certification – **TryHackMe**
- Google Cloud Skills Boost – **Google Cloud**
- Google Professional Cyber Security – **Credly**
- Cybersecurity Roles, Processes & Operating System Security – **Coursera, IBM**
- Introduction of IoT – **NPTEL**
- Introduction to Programming Using Python – **HackerRank**
- Networking Basics – **Cisco**
- Networking Devices and Initial Configuration – **Cisco**